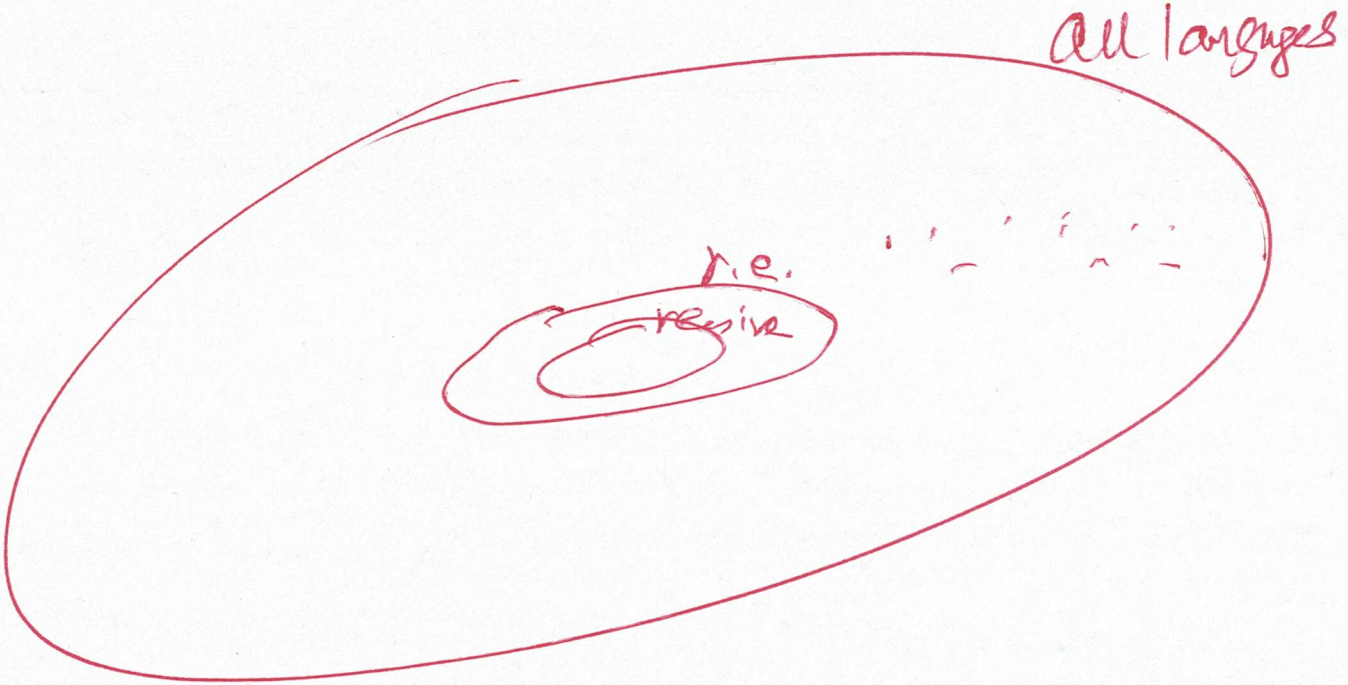
Cpts 317. 12/9.

Key concepts: recursive, r.e., ← languages.

decidable, undecidable ← problems.



Hard: automata-to-automata constructions.

(see your hw).

Example. Given: $\mathcal{L}_1$ —— context-free

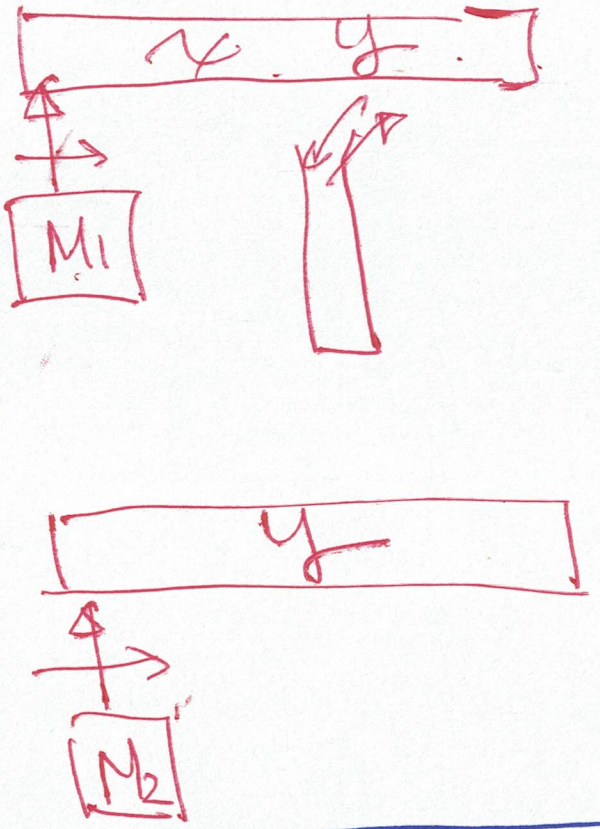
$\mathcal{L}_2$ —— regular.

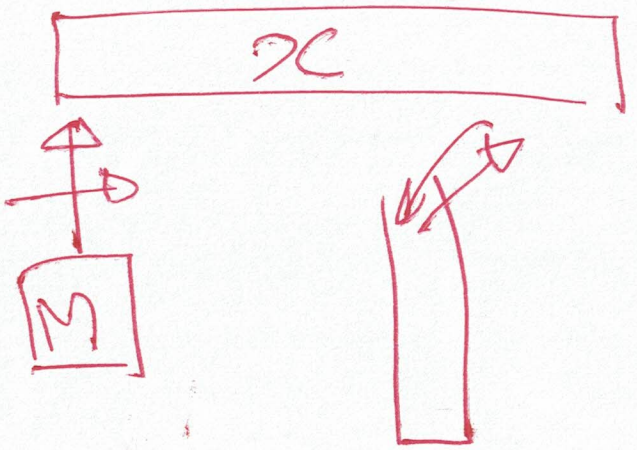
Define

$$\mathcal{L} = \{ x : \exists y : \begin{array}{l} xy \in \mathcal{L}_1 \wedge \\ y \in \mathcal{L}_2 \end{array} \}.$$

Show: $\mathcal{L}$ is c.f.

Proof.





$\exists y.$ M_1 says yes on xy iff M says yes on x

and M_2 says yes on y

Since $\mathcal{L}_1$ is c.f., let M_1 be a PDA accepting $\mathcal{L}_1$. Since $\mathcal{L}_2$ is regular, let M_2 be a FA accepting $\mathcal{L}_2$. We now construct PDA M running on input x as follows. M now simulates PDA M_1 on x. At the end of x, M guesses a word y (symbol by symbol) and lets M_1 continue to run on the guessed y. At the same time, M simulates M_2 running on the guessed y, in parallel (to the run of M_1). At the end of y, M says yes on x if M_1 says yes on xy and M_2 says yes on y. Clearly, M accepts $\mathcal{L}$. $\boxtimes$

4 pages cheating sheets (single sided).

physical paper (no ipad).

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total: 11 problems in 2 hrs.

Rewatch the review videos.

Don't panic!